Does Client Diversity Protect the Lightning Network?

Evidence from the 2026 Core Lightning Shutdown

Saurabh Kumar
Bitcoin Data Labs
sorukumar@bitcoindatalabs.org

***Abstract*—Client diversity is often cited as the Lightning Network's defence against a bug in any one implementation, yet LND runs 80% of public nodes (84% counting one operator's fleet of about 2,000 tiny nodes) and holds 84% of public channel capacity, a capacity share that has barely moved since 2023. We test what diversity buys with a natural experiment. On 26 August 2026, amid security fixes in every implementation, Core Lightning (CLN) maintainers told operators who could not upgrade to disconnect their nodes. Using 805 daily gossip snapshots, 132,360 on-chain channel closes and a client classifier validated against self-identifying aliases and against the wallet that builds each node's funding transactions, we find that 22% of CLN nodes went dark overnight, ten times the usual rate, mostly Tor-reachable ones; a third were still dark five weeks later. The rest of the network barely noticed. Reachability among all node pairs fell by 7.8 percentage points, a loss exceeded in 4 of 94 ordinary two-day windows and half that of a Tor denial-of-service wave in July, while among CLN pairs it fell by 24.5 points, more than in any ordinary window. Calibrated failure scenarios explain the containment: the blast radius depends on whether a failure reaches the hubs, not on how many nodes run the failing software. Losing ACINQ's Eclair node alone costs the surviving network four times the reachability lost with all 619 routable CLN nodes, and minority clients' reliance on LND for connectivity is what their size implies; CLN nodes in fact peer with each other more than chance. Diversity statistics that count nodes mislead; the quantity to track is which software the hubs run.**

***Index Terms*—Lightning Network, Bitcoin, client diversity, common-mode failure, measurement, gossip, resilience**

I. Introduction

The Lightning Network (LN) [1] routes Bitcoin payments over a network of two-party payment channels. Its protocol is defined by the Basis of Lightning Technology (BOLT) specifications [2]–[5], and four independent implementations—LND (Lightning Network Daemon), Core Lightning (CLN), Eclair and LDK (Lightning Development Kit)—interoperate on the same public network. Client diversity is commonly invoked as a safety property: a bug in one implementation should not take down the network. The public network does not look diverse. LND runs 80% of public nodes (84% if one operator's fleet of about 2,000 tiny LND nodes is counted) and holds 84% of public capacity, and its capacity share has stayed between 84% and 90% for three years. Does the remaining diversity protect anything, and is the concentration the risk it appears to be? We study the public (announced) graph, the part of the network that forwards payments for others; wallets on unannounced channels attach to it through their providers' nodes and are not visible in gossip.

Both questions have been hard to answer because client-correlated failures had not been observed. Topological studies [6]–[10] remove random or highly connected nodes from a snapshot; removing "all LND nodes" from a graph in which LND holds 85% of capacity is trivially catastrophic and says little about realistic failures, which are partial, version-scoped and temporary. In 2026 a natural experiment arrived. Between July and October every implementation shipped security fixes, for different bugs, and on 26 August CLN maintainers told operators who could not upgrade to take their nodes off the network. A sizeable cohort of one client went dark at once, in production. We measure what happened and use it to calibrate a model of client-correlated failure.

The answer is that the outage hurt CLN's own users and almost nobody else. The nodes that complied were mostly Tor-reachable nodes at the periphery of the graph, and the network as a whole lost about as much reachability as in an unusual but ordinary two days. What determines the blast radius of a correlated failure is whether it reaches the hubs: a single Eclair node, ACINQ's, matters more to everyone else than the entire CLN cohort, and a transport-level failure (a denial-of-service wave against Tor in July) cost the network twice as much reachability as the CLN shutdown. Counting nodes per client is the wrong way to assess diversity.

Contributions:

- **C1: Validated client inference.** Feature-bit rules that stay valid on historical snapshots are checked against 795 self-identifying aliases and against an independent on-chain signal, the wallet that builds a node's channel funding transactions (§IV).
- **C2: Three years of concentration.** LND's share of public capacity stayed between 84% and 90% from May 2023 to October 2026. Its share of nodes fell by five to seven points from late 2024, a decline that one operator's fleet of small LND nodes hides in the raw counts (§V).
- **C3: An observed client-correlated outage.** We measure the CLN shutdown against churn baselines and a control client: who complied, how long nodes stayed dark, how peers and the closing market reacted, and how the event

differs from transport failures and single-operator closing waves that resemble incidents (§VI).

- **C4: What decides the blast radius.** Calibrated on C3, compared with 94 ordinary two-day windows and with targeted baselines, and tested against a null model of how clients connect, client-correlated failures stay contained; hubs, not cohorts, carry the risk (§VII).

All data are public gossip and blockchain data; code, aggregate datasets and every number quoted in the text are released (see Data and Code Availability).

II. Background and Related Work

Gossip and channel state: Nodes announce themselves (`node_announcement`, including a feature-bit vector [5]) and the routing policy of each channel direction (`channel_update`) [4]. A `channel_update` can mark its direction *disabled*; implementations do so when the peer has been unreachable for some minutes. The volume and propagation delay of such updates have been measured [11]. A node that is offline cannot announce anything; its *peers* disable their direction toward it. Outages are therefore visible as peer-side disabled flags, which is how we measure them. Node availability and reliability have also been studied empirically [12].

Closing channels on-chain: A channel is funded by a 2-of-2 output. A cooperative close spends it with a transaction both parties sign; a unilateral (force) close broadcasts the latest commitment transaction, recognizable by its encoded commitment number in `nLockTime` (upper byte `0x20`) and the input `nSequence` (upper byte `0x80`) [3]. Its outputs are later swept (delayed `to_local`, `to_remote`, anchors, and the outputs of in-flight hash time-locked contracts, HTLCs) or, if a revoked state was published, claimed by the counterparty with a *justice* transaction. Prior work classified closes on-chain [13] and by implementation pair [14], and linked on-chain and gossip data [15], [16].

Resilience and attacks: Rohrer et al. [6], Seres et al. [7], Martinazzi and Flori [9] and Lee and Kim [10] found the LN robust to random failures and fragile to the targeted removal of central nodes; Lin et al. [8] documented its centralisation and showed that removing hubs splits it into many components. Attack studies exploit HTLC handling, congestion, balance locking and correlated mass exits [17]–[20], and implementation defaults shape the cost of some attacks [21]. None conditions failure on the implementation or observes one.

Client inference: Zabka et al. [14], [22] infer implementations from default channel-policy and node-announcement values; Espinasa-Vilarrasa et al. [23] use BOLT #9 feature bits and encrypted traffic metadata. We use feature bits, validate them against an on-chain signal, and keep the rules valid across three years of feature-set changes.

Monoculture and client diversity: That software monoculture concentrates risk is a classic argument in security [24]. Birman and Schneider [25] counter that the risk is often overstated: diversity multiplies the distinct attacks available, and independently built implementations of one specification can share vulnerabilities. In blockchains, client bugs have caused consensus failures, from Bitcoin's 2013 chain fork [26] to Ethereum client incidents [27], motivating differential testing across clients [28] and incentives for diversity [29]. Lightning differs: a client failure takes nodes offline rather than splitting consensus, so its cost depends on where in the payment graph the failing nodes sit, which is what we measure.

III. Data and Vantage Point

All inputs are truncated at a data cutoff of 4 October 2026.

Daily gossip: Our LND node dumps its view of the public graph (`describegraph`) daily. Between 8 May 2023 and the cutoff (1,246 calendar days), 857 dumps produced usable extracts; the other days are missing, mostly in 2023 and early 2025, or were error responses from LND ("wallet locked" after restarts). A quality pass excluded 51 partial dumps (May–June 2024) and one stale day, leaving 805 days (65% of the calendar). Coverage is complete enough for daily analysis from May 2026, the window of §VI. The node was replaced in mid-2024, which changed the view: before, about 42k channels and 8.4k nodes with channels; after, 32–48k channels (median 41k) and 10–13k nodes. We treat the two periods as *vantage epochs* and compare levels only within an epoch. Gossip is a single node's view of the public (announced) graph; unannounced channels, where much LDK (wallets) and Eclair (Phoenix) usage lives, are not visible.

On-chain closes and spends: Our own Bitcoin Core node scans every block since January 2025 for spends of the 257,363 funding outputs of every channel ever seen in our gossip. For each close we record the transaction, block time and type (cooperative, force, splice); for each output of a force close, the transaction that spends it and its type from the witness script. The data contain 132,360 closes (99,016 cooperative, 33,321 force, 23 splices) and 112,837 spends of force-close outputs. Batched sweeps' fees are split across their inputs.

Funding transactions: For the 226,134 funding transactions of channels in our catalogue we record the funder wallet's construction choices: `nVersion`, `nLockTime` policy, input `nSequence`, input and change script types, output ordering, signature grinding, fee rate.

IV. Client Inference and Validation

Rules: Implementation-specific feature bits decide most nodes. Bits 2023 (script-enforced lease) and 31 (AMP, atomic multi-path payments), which LND assigns outside BOLT #9, appear only on LND. Bit 37 (`option_attribution_data`), or 29 (`option_dual_fund`) together with 61 (`option_simple_close`), marks Eclair. Bit 11 (`gossip_queries_ex`) together with 39 (`option_onion_messages`) or 43 (`option_provide_storage`) marks CLN or Eclair, which we separate by bit 55 (keysend): CLN advertises it by default (about 96% of CLN-set nodes), Eclair does not. Bits 39, 43 or 63 (`option_splice`) without 11

mark LDK. Fallbacks (LND's default colour and alias, and the modal default CLTV delta, the number of blocks of CheckLockTimeVerify time-lock a node requires to forward a payment) yield low or medium confidence. Operators with a publicly known implementation (ACINQ, Block) are attributed directly. An earlier version without the keysend rule labeled ACINQ as CLN before 2025, when Eclair added bits 37 and 61: rules must be validated against historical snapshots, not only the latest one.

On-chain signal: A funding transaction is built by the funder's wallet. On channels whose two endpoints share a high-confidence label (so the funder's software is known), two wallet families separate cleanly: LND's btcwallet (`nLockTime` 0, `nSequence 0x0`) and Core/BDK-style wallets (anti-fee-sniping `nLockTime` at the current height, `nSequence 0xfffffffd`), used by CLN, Eclair (via bitcoind) and software based on the Bitcoin Development Kit (BDK). Since chain data do not reveal which endpoint funded a channel, we score each node with a likelihood ratio in which a channel is funded by the node with probability $\theta = 0.5$ and otherwise by the peer, whose wallet distribution follows the peer's label; batch opens, where one node is common to every channel funded by the transaction, identify the funder exactly. The signal measures the *wallet*, not the daemon: LND operators who fund channels from an external wallet look Core-style. It separates LND from the other clients but cannot tell CLN from Eclair or identify LDK (our LDK sample is 99% one operator).

Validation: Table I and Fig. 1 report two independent checks. Against 795 nodes whose alias names their implementation (pooled over monthly snapshots; LND's default alias excluded because the rules use it), precision and recall are 99.5%/94% for LND and 97%/96% for CLN. Most LND misses are nodes without announced features, which the rules leave Unknown. Eclair (9 nodes, precision 62%) and LDK (6) are too few to grade; Eclair-labeled nodes that lack bit 37 may be CLN nodes with keysend disabled, so Eclair's node count is an upper bound, although its capacity is dominated by ACINQ, which is attributed directly. Against the on-chain wallet family (5-fold held out for high-confidence labels), nodes labeled by LND-only bits agree 99% of the time (n = 3,199) and the CLN set with keysend 98% (n = 471); the CLTV-only fallbacks are weakest (83–95%) and account for five of the nine alias disagreements (e.g. LND nodes of one node-in-a-box platform use CLTV 144, Eclair's default). A learned baseline (k-nearest neighbours on the feature-bit vector) does not beat the rules on the same held-out folds (macro-F1 0.92 vs. 0.93). Over all nodes with at least five funding transactions, held-out agreement between the on-chain family and high-confidence labels is 99.2–99.3%, and 98% of nodes get a decision; because the signal grades LND against everything else and most nodes are LND, the per-tier figures above are the more informative ones.

TABLE I

VALIDATION OF CLIENT INFERENCE (RULES V3-2026-10-06). (A) 795 NODES WHOSE ALIAS NAMES THEIR IMPLEMENTATION (POOLED OVER MONTHLY SNAPSHOTS, EVALUATED ON THE LATEST SNAPSHOT WHERE THEY HAVE A CHANNEL); "UNKNOWN" COUNTS NODES THE RULES LEAVE UNLABELED (MOSTLY NO ANNOUNCED FEATURES). A LEARNED BASELINE (k-NN, JACCARD, ON THE FEATURE-BIT VECTOR) VS. RULES ON THE SAME HELD-OUT FOLDS: MACRO-F1 0.92 VS. 0.93 ($n = 761$). (B) SHARE OF NODES IN EACH RULE TIER WHOSE CHANNEL-FUNDING WALLET AGREES WITH THE LABEL (LND BTCWALLET FOR LND, CORE/BDK-STYLE OTHERWISE), NODES WITH DECISIVE ON-CHAIN EVIDENCE; HIGH TIERS GRADED WITH HELD-OUT EVIDENCE.

(a) Alias anchors

Client	n	Precision (%)	Recall (%)	Unknown
LND	664	99.5	94.0	35
CLN	116	97.4	95.7	2
Eclair	9	62.5	55.6	3
LDK	6	100.0	100.0	0

(b) On-chain wallet family

Label	Rule tier	n	Agree (%)
CLN	CLN set + keysend	471	98.1
CLN	CLTV default only	38	94.7
CLN	CLTV default only (no features)	18	83.3
Eclair	Eclair bits (37, 29+61)	10	90.0
LDK	CLTV default only (no features)	13	100.0
LND	LND-only bits (2023/31)	3,199	99.2

V. CONCENTRATION, 2023–2026

Figure 2 shows monthly client shares of public nodes and of node capacity (each channel counted for both endpoints). LND's share of capacity stayed between 84% and 90% throughout. Its share of nodes fell from 91.8% (May 2023) to 84.2% (October 2026). The step between epochs coincides with the change of our gossip node and the appearance of many featureless (Unknown) nodes, not with a change in the network; within the first epoch the share moved by about one point (91.8% to 90.9%). Within the second, the raw share looks stable (84.2% at the end against 85.1% at the start), but one operator props it up. From July 2025 a managed fleet of small LND nodes with aliases `LNT.<name>`, each with one or two channels, mostly to the operator's own hubs, grew to 1,980 nodes (March 2026), a fifth of all public nodes with channels; on the latest snapshot its 1,972 nodes hold 1.4 BTC (0.02% of capacity). We count a node as part of the fleet if its alias carries the prefix or at least half of its channels go to a fleet hub. Without it, LND's share of nodes fell steadily, from 85–87% in the second half of 2024 to 80.3% in October 2026. In capacity, the only clear within-epoch trend is LDK's share, from 1.7% (July 2024) to 5.4%, mostly Block's routing nodes. On the latest snapshot (Table II) LND runs 84.0% of nodes, 80.1% without the fleet (77.5% and 71.9% counting only high-confidence labels), and holds 84.4% of capacity; CLN runs 8.5% of nodes (10.6% without the fleet) but holds 3.7% of capacity; Eclair's 43 nodes hold 6.1%. Capacity is highly unequal within every cohort (Gini 0.92–0.97). The largest nodes are, if anything, more concentrated by client than nodes in general: LND runs 90 of the 100 largest

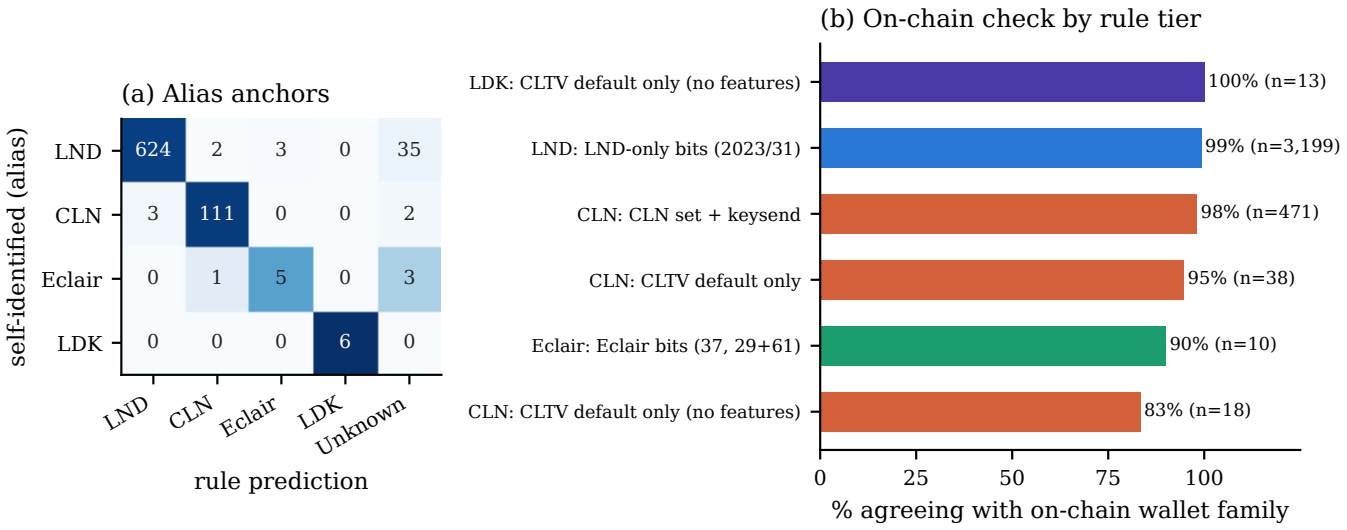


Fig. 1. Validation of client inference. (a) Nodes whose alias names their implementation vs. the rule label. (b) Share of nodes in each rule tier whose channel-funding wallet agrees with the label (tiers with $n \geq 10$).

nodes of the routable core (§VII) and 8 of the 10 largest; the other two are ACINQ's Eclair node and Block's LDK node.

VI. The 2026 Security Wave and the CLN Shutdown

Timeline: On 19 July a researcher disclosed two memory-exhaustion denial-of-service bugs in CLN, already fixed [30]. On 7 August BTCPay Server reported that its versions before 2.4.2 let remote attackers obtain the LND credentials (macaroons) of merchant nodes and that funds had been stolen [31]; affected merchants found channels force-closed and balances swept [32]. This was an application-layer credential leak, not an LND protocol bug. On 11 August Lightning Labs published ten LND advisories, three of high severity, all for versions before 0.19 [33]. On 26 August CLN maintainers, after weeks of triaging a high volume of vulnerability reports, advised operators to upgrade or, if they could not, to restart with `--offline`, which keeps the node running and watching the chain but drops every peer connection; the advice circulated during the day and the project posted it publicly at 23:26 UTC [34], [35]. CLN 26.06.7 shipped on 28 August [36], LDK 0.2.6 on 9 September [37] and Eclair 0.14.3 on 14 September [38]. Lightning Labs published eight more advisories (four for LND, four for btcd) on 21 September [39], CLN 26.06.8 followed on 22 September [40], and on 2 October CLN warned that attackers were targeting nodes running 26.06.7 or earlier [41]. Every implementation was affected; the bugs were different.

Measure: For client c and day t, $D_c(t)$ is the share of channel directions pointing *toward* nodes of c that their peers mark disabled, unweighted and weighted by capacity. A node restarted with `--offline` and a node that was shut down look the same to its peers. Client labels are fixed per node on 1 August 2026, so the incident cannot reshuffle cohorts. The baseline is 1–25 August.

Magnitude: On 27 August D_{CLN} rose from a baseline of $11.8 \pm 0.9\%$ to 38.1% ($z = 29$); by capacity from 7.4% to 35.2% (Table III, Fig. 3). LND, Eclair and LDK did not move ($|z| < 2$). A difference-in-differences estimate against LND over the following two weeks, with a moving-block bootstrap, gives $+14.2$ percentage points (95% CI 11.2–16.0) by channel count and $+14.7$ (10.4–17.1) by capacity. Removing, each day, the single node with the most disabled capacity leaves $+12.9$ (8.9–15.2), so the capacity result is not one large node (Table V).

Who complied: We call a node *dark* when at least 80% of the channel directions toward it are disabled, or when it is absent from gossip. Of 624 CLN nodes not dark on 25 August, 137 (22.0%) were dark on 27 August, holding 33% of CLN capacity. On an ordinary day the same transition (not dark on day $t-1$, dark on $t+1$) affects 2.3% of CLN nodes (median of 97 days, May–August; 95th percentile 5.6%, maximum 12.3%), so the call took about ten times the usual number of CLN nodes off the network. LND nodes over the same days went dark at 6.4%, within their ordinary range (median 5.2%, 95th percentile 9.5%). Larger operators complied more: 32% of the largest capacity quintile against 8–27% of the others (Table IV). Compliance split by how nodes are reachable: 28% ($n = 310$) of Tor-only CLN nodes went dark, 21% ($n = 150$) of dual-stack nodes and 7% ($n = 90$) of clearnet-only nodes. Toward Tor-only CLN nodes the disabled share rose from 15.7% to 39.0%, toward dual-stack nodes from 6.9% to 47.4%, while toward clearnet-only nodes it rose from 9.7% to 19.0% for two days and the two-week difference-in-differences is zero (Table V). Tor-only LND nodes did not move (25.6% baseline, 23.4% on 27 August), so this was not a Tor outage. A plausible reading, which our data cannot confirm, is that

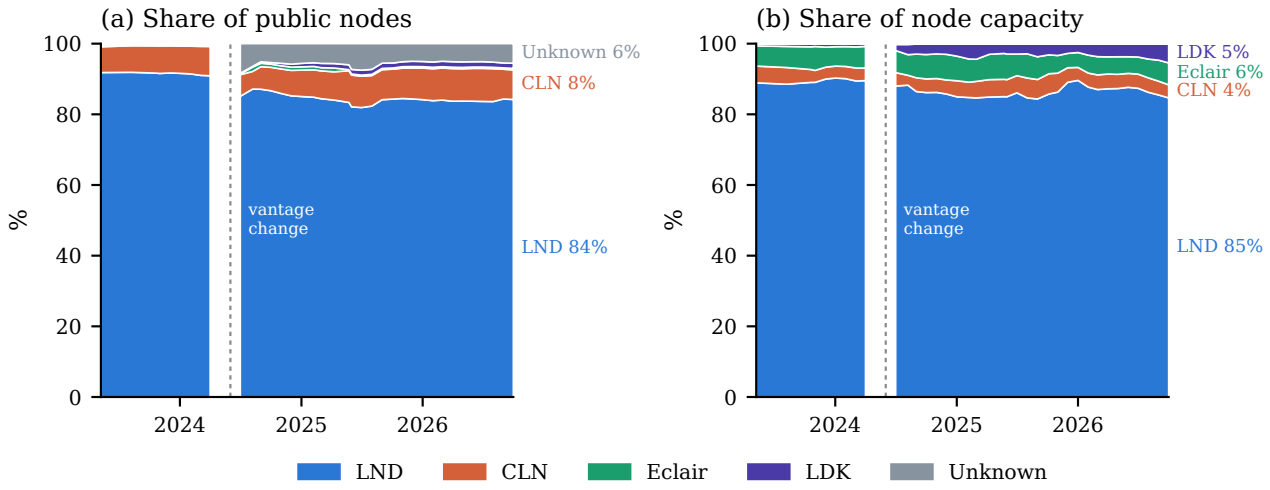


Fig. 2. Client shares of public nodes with at least one channel (a) and of node capacity (b), first gossip snapshot of each month. The dashed line marks the change of our gossip vantage point (mid-2024); months with unusable dumps are omitted. From mid-2025 the LND node share includes one operator's fleet of up to 1,980 small nodes (§V).

TABLE II

CLIENT IMPLEMENTATIONS OF PUBLIC LIGHTNING NODES WITH AT LEAST ONE CHANNEL, GOSSIP SNAPSHOT 2026-10-04 (RULES V3-2026-10-06). SHARES IN %; IN PARENTHESES, THE SHARE COUNTING ONLY HIGH-CONFIDENCE LABELS (LOWER BOUND). NODE CAPACITY COUNTS EACH CHANNEL FOR BOTH ENDPOINTS (TOTAL 7,390 BTC). GINI: INEQUALITY OF NODE CAPACITY WITHIN THE COHORT.

Client	Nodes	Node share	Endpoints	Capacity (BTC)	Cap. share	Mean (BTC)	Gini
LND	8,290	84.0 (77.5)	85.6	6,237	84.4 (84.3)	0.75	0.97
CLN	834	8.5 (7.2)	7.8	276	3.7 (3.7)	0.33	0.92
Eclair	43	0.4 (0.1)	3.6	452	6.1 (5.6)	10.52	0.95
LDK	148	1.5 (1.1)	2.0	422	5.7 (5.7)	2.85	0.97
Unknown	553	5.6 (0.0)	1.0	3	0.0 (0.0)	0.01	0.68

TABLE III

SHARE OF CHANNEL DIRECTIONS TOWARD EACH CLIENT COHORT DISABLED BY PEERS (%). BASELINE: DAILY MEAN ± SD, 2026-08-01 TO 2026-08-25; PEAK: 2026-08-27; z = (PEAK − MEAN)/SD. DIFFERENCE-IN-DIFFERENCES, CLN VS. LND, 2026-08-27 TO 2026-09-09 VS. BASELINE (95% MOVING-BLOCK BOOTSTRAP CI, BLOCK 7 DAYS): CHANNELS: +14.2 PP [+11.2, +16.0]; CAPACITY: +14.7 PP [+10.4, +17.1].

	By channel count			By capacity		
Client	Baseline	Peak	z	Baseline	Peak	z
LND	16.4 ± 0.8	16.6	0.3	3.5 ± 1.6	2.4	-0.7
CLN	11.8 ± 0.9	38.1	29.4	7.4 ± 1.8	35.2	15.5
Eclair	3.2 ± 0.2	3.5	1.2	0.8 ± 0.6	1.7	1.5
LDK	7.7 ± 0.7	6.6	-1.7	1.6 ± 0.6	1.3	-0.6

home and node-in-a-box operators, typically on Tor, followed the call, and hosted operators upgraded instead.

A transport-level contrast: Gossip also shows a common-mode failure that was *not* client-correlated. On 10 July the disabled share toward Tor-only nodes of every client rose from 23.7% to 43.5% overnight (clearnet-only nodes flat) and decayed over two weeks (Fig. 4). It coincides with a denial-of-service wave against the Tor network, in which the number of relays serving onion-service descriptors fell from 3,874 on 9 July to 1,780 on 12 July and recovered only after 22 July [42]–[44]; Tor-only Lightning nodes are reachable only through those descriptors. Common-mode failures follow transport as well as software.

Recovery: Of the 137 nodes, 39% were back within three days and 53% within a week; among nodes that returned, the median time dark was three days (interquartile range 3–7), so most returns came shortly after the release of CLN 26.06.7 on 28 August, consistent with operators upgrading and reconnecting (gossip does not reveal versions). Twenty-two percent never reappeared, even for a day, by 4 October; counting nodes that came back and went dark again, 32% were dark on that date (Fig. 5). The 2 October warning produced no comparable second wave (D_{CLN} 14.6% on 2 October, 15.3% on 4 October).

Closing behaviour and confounds: Read naively, closing data are full of "incidents". The two largest force-close days in the window were single operators: on 3 August one LND node (NiceHash) was party to 226 force closes, and on 26 August—the day of the CLN call—another LND node (Wallet of Satoshi), whose operator had announced a move away from its custodial Lightning wallet [45], was party to 205. The early-August cooperative-close wave was driven by large operators

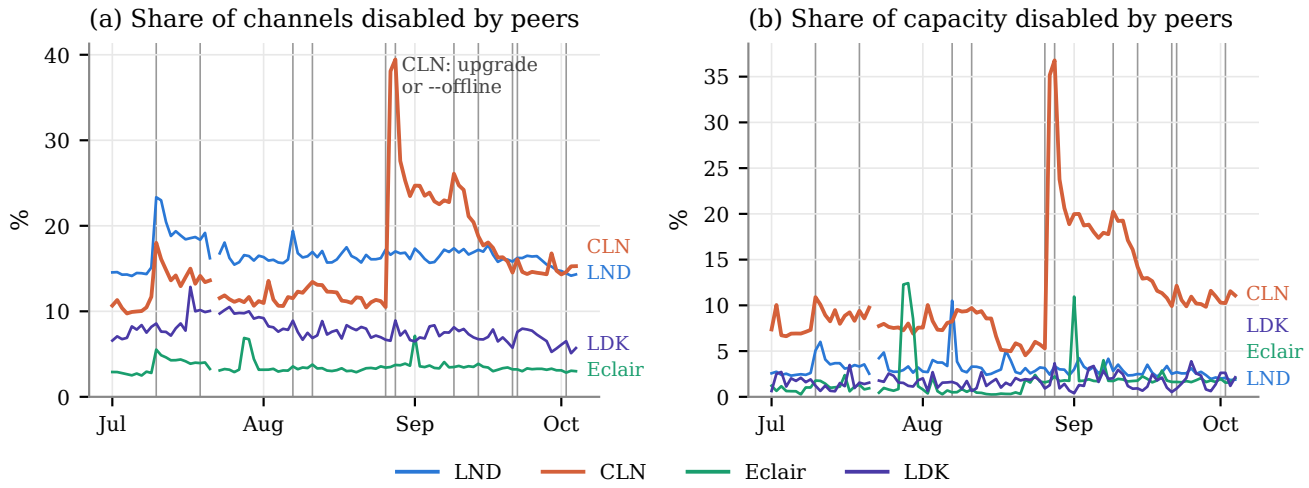


Fig. 3. Share of channel directions toward each client's nodes that peers mark disabled, daily, July–October 2026; by channel count (a) and by capacity (b). Vertical lines: events of §VI.

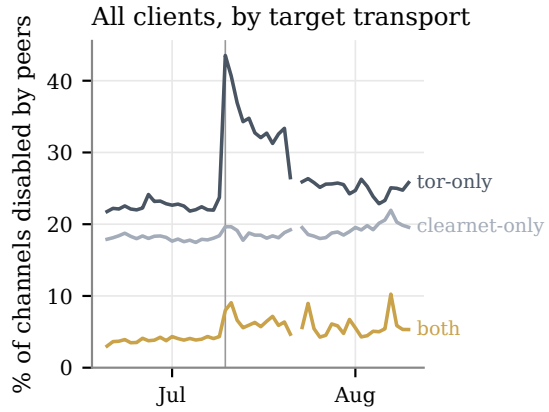


Fig. 4. Disabled share toward Tor-only, clearnet-only and dual-stack nodes, all clients pooled. On 10 July 2026, during a denial-of-service wave against Tor, Tor-only nodes of every client lost their peers at once.

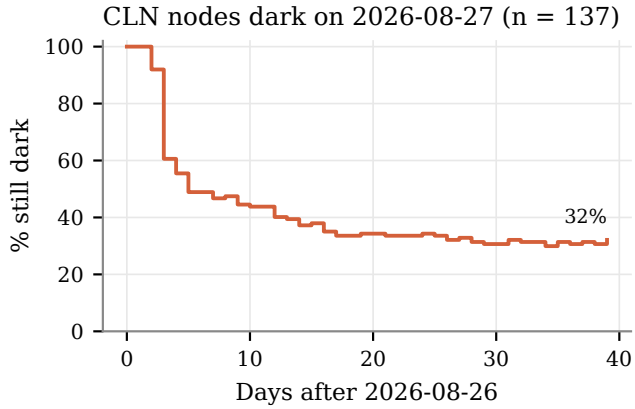


Fig. 5. Share of the CLN nodes that went dark after the shutdown call that were still dark, by day.

closing hundreds of channels each: Kraken, which moved to a new node on 12 August [46], was party to 12.6% of all closes on 2–16 August. We therefore separate *mass-close events*, a node party to at least 20 force closes in a UTC day (37 events since May, 30 by LND nodes), from organic force closes. Organic force closes of channels involving a CLN node ran at 2.2 times their July–August rate during 26 August–8 September, 2.6 times relative to channels between two LND nodes, whose rate fell slightly (0.83 times; Table IV). Within CLN, the channels of nodes that went dark were force-closed at 70 per thousand channels in those two weeks, against 43 for CLN nodes that stayed up. An `--offline` node can still broadcast its own commitment transactions, and chain data do not reveal which side broadcast a close, so we cannot attribute these closes to peers. Force closes are not a sign of theft: we find 17 justice spends in 21 months, the last on 11 July 2026; a successful theft leaves none, so this bounds attempts only from below.

A contracting network: The security wave coincided with the largest contraction of the public graph in our data, and the contraction was not client-specific. Of the 38,081 channels in gossip on 1 August, 14,547 (38%, holding 44% of the capacity) were gone by 4 October; 97% of them closed on-chain, 10,464 in August, against 6,833 new channels. In the window of the same length starting 1 May, 18% of channels disappeared. Channels involving LND, CLN, Eclair and LDK nodes disappeared at similar rates (38%, 42%, 32% and 48%); 79% of the closes were cooperative, and ten operators, led by Kraken's node migration, were party to 31% of them. Exits by large operators, not client failures, account for the network's shrinkage.

Cost: The median force close since May cost 1,222 sat in fees including the sweeps of its outputs, six times a cooperative

TABLE IV

(A) CLN NODES THAT WENT DARK AFTER THE SHUTDOWN CALL (NOT DARK ON 2026-08-25, DARK ON 2026-08-27), BY PRE-INCIDENT NODE CAPACITY QUINTILE. (B) ORGANIC FORCE CLOSES PER 1,000 OPEN PUBLIC CHANNELS PER DAY INVOLVING EACH COHORT, 2026-07-01–08-25 (BASELINE) VS. 2026-08-26–09-08 (WINDOW); MASS-CLOSE EVENTS (ONE NODE IN $\geq$ 20 FORCE CLOSES IN A UTC DAY) EXCLUDED.

(a) Compliance by size

Quintile	Capacity (BTC)	Nodes	Went dark (%)
Q1	0.000–0.004	125	23.2
Q2	0.004–0.010	125	8.0
Q3	0.010–0.041	124	19.4
Q4	0.042–0.205	125	27.2
Q5	0.208–44.488	125	32.0

(b) Organic force-close rate

Cohort	Baseline	Window	Ratio	vs. LND↔LND
CLN	1.40	3.07	2.18	2.63
Eclair	0.79	0.84	1.06	1.28
LDK	14.17	8.45	0.60	0.72
LND↔LND	1.02	0.85	0.83	1.00

close (209 sat); all 8,787 force closes paid 0.17 BTC in total. The on-chain cost of the episode was negligible; its cost was liquidity locked behind time-locks and channels lost.

VII. WHAT DECIDES THE BLAST RADIUS

Model: We take the routable core of the public graph on 25 August 2026, the day before the call: channels whose two directions are announced and enabled (73% of announced channels; 4,435 nodes, of which 189 belong to the fleet of §V, almost all with a single channel). A payment of a satoshis can use a channel whose capacity is at least a (optimistic) or $2a$ (balances split evenly). Failures are modelled as *exposure*, a fraction p of a client's nodes being unavailable, not deletion of the client. Our main measure is $R_{\mathrm{all}}(a)$, the share of the pre-failure node pairs still connected by a path of usable channels, so pairs involving a failed node count as lost; we also report $R(a)$ among surviving nodes and $R_c(a)$ among surviving pairs of client c. Connectivity is an upper bound on what routing achieves (no fees, no in-flight HTLCs, no liquidity uncertainty beyond the threshold). Random scenarios average 50 draws. Eclair has only 12 nodes in the core, so R_{Eclair} moves in large steps.

The observed footprint against ordinary churn: The observed footprint is the set of nodes of the 25 August core with no usable channel left on 27 August (383 nodes, 105 of them CLN). It reduced R_{all} for 100k-sat payments by 7.8 points. We compute the same two-day footprint for every pair of snapshots two days apart from May to August (Fig. 6). In 94 windows without a known incident the loss had a median of 3.6 points; 4 of them lost more than the CLN shutdown. For pairs of CLN nodes, counting pairs with a failed member as lost, the outage cost 24.5 points; no ordinary window cost more than 17.7 (median 3.2). The two windows spanning the Tor event of 10 July cost the whole network 15.6–16.0 points, twice the CLN shutdown.

Scenarios: Table VI and Fig. 7 compare failures. The calibrated scenario ($\hat{p} = 0.22$ of CLN nodes failing at random, 136 nodes) leaves R among survivors at 78.8%, against 77.9% when the same number of nodes fails at random across all clients: for the rest of the network, a CLN-wide outage behaves like random failure, because CLN nodes are mostly peripheral. Drawing the failures with the observed compliance of each size quintile and transport class instead gives the same result ($R = 78.7\%$, $R_{\mathrm{all}} = 73.9\%$). The damage stays inside the cohort: counting pairs with a failed member as lost, CLN pairs fall from 79% to 46% connected, while among surviving CLN nodes R_{CLN} is 76%. Removing the same number of nodes chosen by degree leaves 27.6%, by capacity 33.7%. Losing all 619 CLN nodes of the core leaves survivors at 76.2%; losing its 12 Eclair nodes, which hold 5.8% of capacity and are dominated by ACINQ, leaves 63.9%. Per unit of failed capacity, failures confined to Eclair are the most damaging and failures confined to LDK the least (Fig. 7).

Hubs, not cohorts: Single nodes explain the gap between cohorts. Removing each of the 300 largest nodes of the core on its own, the most critical by far is ACINQ's Eclair node: it holds 4.9% of node capacity, and its loss alone cuts R by 15.9 points, four times the loss from all 619 CLN nodes (3.7 points). The next most critical nodes, a Lightning service provider and a payment processor running LND, cost 2.8 and 1.9 points; the 20 most critical nodes, 19 of them LND, cost a median of 0.6 points each. Criticality follows a node's role rather than its size: the two largest nodes by capacity, both exchanges, are not among the six most critical, whereas nodes that are the only adequate link for many small nodes are. The software of these few nodes, not the client shares of Table II, decides how far a client-correlated failure reaches.

Dependence on LND, against a null model: Minority clients' channel capacity faces LND: 76% for CLN, 92% for Eclair and 81% for LDK (Fig. 8), and without LND nodes only 32% of CLN pairs, 42% of Eclair pairs and 11% of LDK pairs remain connected for 100k-sat payments (14%, 42% and 9% for 1M sats). Read alone, this suggests that minority clients reach each other only through LND. A null model says otherwise. Permuting client labels among nodes of similar capacity (within capacity deciles, 200 draws) keeps the graph and the cohort sizes and removes any preference of a client for its own kind. Under it, CLN would put 88% of its capacity on LND peers and LDK 87%; both put less. CLN nodes keep 15% of their capacity among CLN peers against 8% by chance, and LDK 15% against 2%. Without LND, 9% of CLN pairs would remain connected by chance (95th percentile 25%), against 32% observed; for Eclair and LDK the observed values lie within the null range. Minority clients' dependence on LND is what their size implies; CLN operators in fact build more of their own backbone than chance would, but it is too small to stand alone.

VIII. DISCUSSION

Containment, not collapse: The first measured client-wide outage in the LN was absorbed. The network as a whole

TABLE V
ROBUSTNESS OF THE CLN OUTAGE EFFECT. BASELINE AND PEAK AS IN TABLE III; DID VS. LND TARGETS IN THE SAME SUBSET, 95% MOVING-BLOCK BOOTSTRAP CI. TARGET TRANSPORT FROM NODE ANNOUNCEMENTS ON 2026-08-01.

Variant	Baseline (%)	Peak (%)	z	DiD (pp)
All targets, by channels	11.8	38.1	29.4	+14.2 [+11.2, +16.0]
All targets, by capacity	7.4	35.2	15.5	+14.7 [+10.4, +17.1]
By capacity, largest node removed	6.1	31.8	20.6	+12.9 [+8.9, +15.2]
Clearnet-only targets, by channels	9.7	19.0	9.3	-0.0 [-1.5, +1.6]
Tor-only targets, by channels	15.7	39.0	9.4	+23.1 [+20.8, +25.2]
Both targets, by channels	6.9	47.4	39.6	+16.1 [+10.0, +18.7]

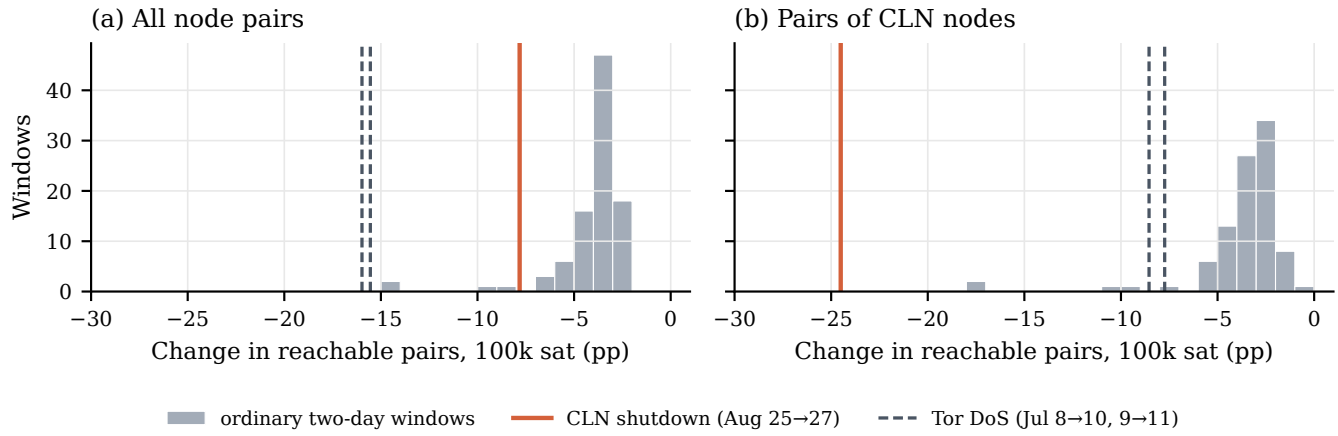


Fig. 6. Change in the share of pre-failure node pairs that can still route 100k sat, from each gossip snapshot to the one two days later (nodes that lost every usable channel count as failed), May–August 2026. Grey: windows without a known incident; solid line: 25→27 August (CLN shutdown); dashed: 8→10 and 9→11 July (Tor denial-of-service wave). (a) All node pairs; (b) pairs of CLN nodes.

lost about as much reachability as in an unusual but ordinary two days, and half as much as in the Tor denial-of-service wave a month earlier, while the affected cohort lost a third of its internal reachability and saw organic force closes double. Client-correlated failure was costly for the client's own users, not for everyone else, because the nodes that complied were peripheral.

Where the risk is: The blast radius of a correlated failure is set by its overlap with the hubs. A few large nodes, regardless of client, carry the network's reachability; ACINQ alone outweighs the entire CLN cohort for the survivors. Node-count diversity statistics therefore mislead in both directions: CLN's 8.5% of nodes protected little, and LND's 84%, four points of which are one operator's fleet, matters mainly through the subset of LND nodes that are hubs. The structure that diversity statistics hide is not a pathological dependence of minority clients on LND, which is what their size implies, but the concentration of connectivity in a few operators, whose software choice is a single point of failure for everyone.

Operational lessons: Compliance with an emergency call was voluntary and uneven (22% of CLN nodes, mostly Tor-reachable); a third of those were still dark after five weeks, so emergency shutdowns have lasting costs for the operators who follow them. Advising `--offline` rather than a shutdown was the safer design, because an offline node keeps watching the chain for revoked states and expiring HTLCs. Single-operator events (mass closes, an exchange migrating its node, a custodial wallet winding down) are easily mistaken for network-wide effects; concentration checks belong in any incident analysis. Gossip pipelines should validate dumps (we found saved error responses) and record their vantage point.

Recommendations: (i) Monitor peer-side disabled shares per client and per transport as an early-warning signal; it is cheap and was unambiguous for both the Tor and the CLN events. (ii) Report client diversity weighted by hub capacity or reachability, not by node count. (iii) Encourage channels between minority-client hubs, so that CLN, Eclair and LDK users are not connected only through LND. (iv) Scope emergency guidance by version and duration, and prefer `--offline` to shutdown.

IX. LIMITATIONS

Our gossip is one node's view of the announced graph, which changed in mid-2024 and has gaps before 2026. Unannounced channels (most LDK and Phoenix usage) are invisible; their users reach the network through their providers' nodes (Phoenix users through ACINQ's), so hub failures cost payments more than §VII measures. Feature-bit rules can be wrong for new releases and are weak where nodes announce no features; Eclair's node count is an upper bound. Node counts treat every node as an independent operator; we found

TABLE VI

FAILURE SCENARIOS ON THE 2026-08-25 PUBLIC GRAPH (CHANNELS ENABLED IN BOTH DIRECTIONS), PAYMENT SIZE 100,000 SATS, OPTIMISTIC LIQUIDITY. R_{all}: % OF PRE-FAILURE NODE PAIRS STILL CONNECTED (PAIRS WITH A FAILED NODE COUNT AS LOST); R: % OF ONLINE NODE PAIRS CONNECTED BY A PATH OF CHANNELS THAT CAN CARRY THE PAYMENT; R_c: SAME AS R, FOR PAIRS OF SURVIVING NODES OF CLIENT c; LCC CAP.: % OF USABLE CAPACITY IN THE LARGEST COMPONENT. RANDOM SCENARIOS: MEAN OF THE MONTE CARLO RUNS.

Scenario	Failed	Cap. (%)	R_{all}	R	R_{CLN}	R_{Eclair}	R_{LDK}	LCC cap.
Intact	0	0.0	79.8	79.8	79.2	83.3	80.0	100.0
Observed: lost all usable channels, Aug 27	383	0.5	72.0	86.3	79.4	83.3	81.5	99.0
of which CLN nodes only	105	0.2	76.0	79.8	79.4	83.3	80.0	99.6
Calibrated CLN ($\hat{p}$ = 0.22)	136	0.9	74.1	78.8	76.3	83.3	79.1	98.2
Same count, random nodes	136	3.1	73.2	77.9	76.8	82.5	77.6	93.9
Same count, top degree	136	70.5	25.9	27.6	24.8	33.3	14.7	8.9
Same count, top capacity	136	80.2	31.7	33.7	30.1	21.4	10.4	6.6
All LND nodes fail	3,704	85.4	0.8	28.3	31.8	42.4	11.1	2.3
All CLN nodes fail	619	3.9	56.4	76.2	–	83.3	76.2	92.8
All Eclair nodes fail	12	5.8	63.5	63.9	59.1	–	69.0	88.4
All LDK nodes fail	95	4.8	76.2	79.6	78.9	83.3	–	91.1

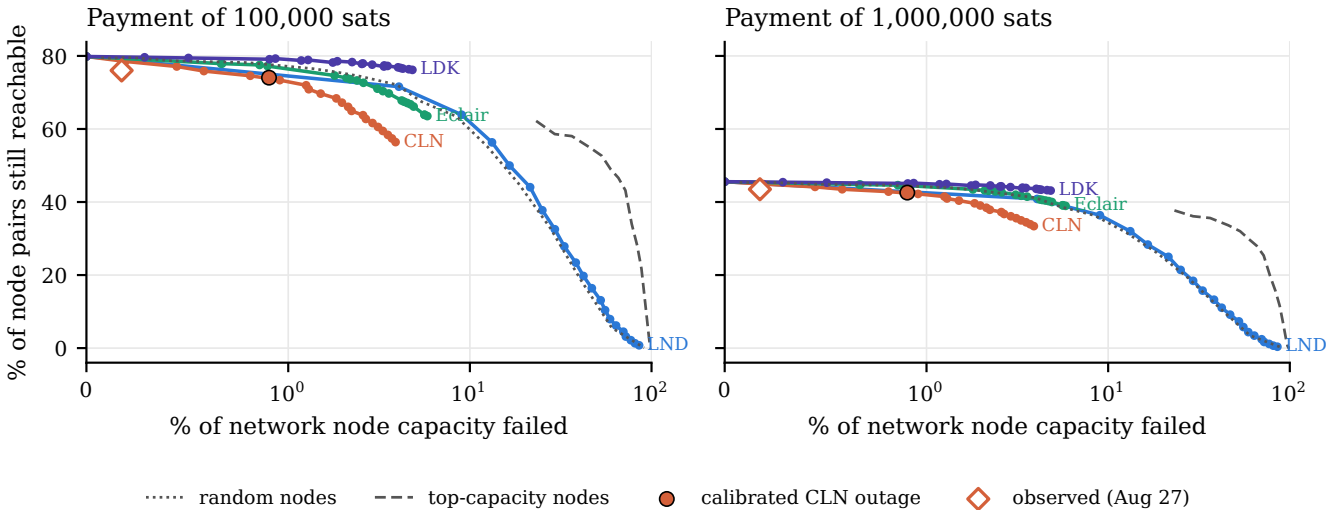


Fig. 7. Share of node pairs still reachable vs. share of network node capacity failed, for random failures within each client (lines with markers), random failures across all nodes (dotted) and the largest nodes by capacity (dashed). Dot: calibrated CLN outage; diamond: observed footprint of 27 August (CLN nodes). Routable core of 25 August 2026; left: 100k sat, right: 1M sat payments.

one managed fleet by its alias and hubs, and smaller fleets without a shared alias would go undetected. The on-chain signal identifies wallets, not daemons, and cannot separate CLN from Eclair or identify LDK. Disabled flags record peers' views with some delay, and a single vantage point can miss or delay updates; such gaps are not specific to one client, and LND, Eclair and LDK, observed through the same node, did not move on 27 August. A channel between two CLN nodes that both went dark is never disabled, so D_{CLN} understates the outage. The dark threshold (80%) and the two-day footprint are choices; the churn baselines and placebo windows use the same definitions. The resilience model measures connectivity under static capacity thresholds, an upper bound on payment success. Channel balances are private, so we do not model how payments rerouted around failed nodes drain liquidity on the surviving paths, nor fees, HTLC limits or private routing hints. The absolute reachability levels therefore overstate what payments achieve; the observed outage, the placebo windows and every scenario are measured with the same model, so the comparisons between them are like for like. Chain data do not reveal which side broadcast a force close. Event dates are those of public announcements; some advice circulated earlier in community channels.

X. ETHICS

We use only public gossip and blockchain data and report aggregates. We name large public operators only for their closing behaviour, which is public on-chain and in the operators' own announcements, and for ACINQ's widely known role as a hub; we describe other critical nodes by type and publish no lists of potentially vulnerable nodes.

XI. CONCLUSION

LND holds about 85% of public Lightning capacity, a share that has barely moved in three years. When an emergency took

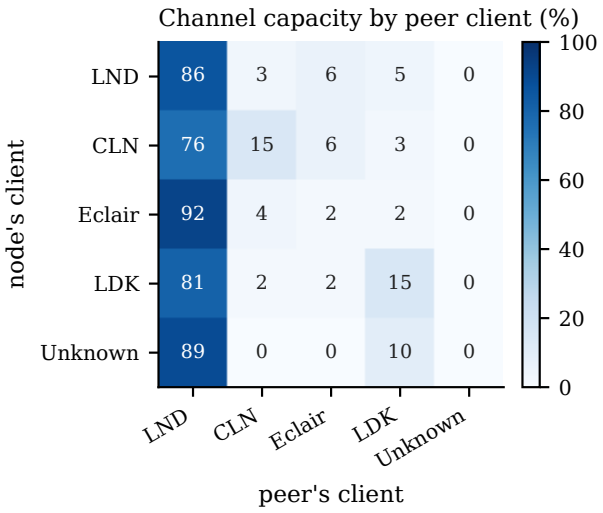


Fig. 8. Share of each client's channel capacity (rows) by the client of the peer (columns), routable core of 25 August 2026.

a fifth of CLN nodes offline in August 2026, the network absorbed it: the damage stayed within the CLN cohort, and a transport failure a month earlier had cost the network twice as much. The fragility of the network lies not in how many nodes run each client but in which software its few hubs run. Client diversity protects the Lightning Network to the extent that it reaches those hubs.

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Data and Code Availability

Code, the aggregate datasets behind every figure and table, and a script that recomputes every number quoted in the text are available at https://github.com/bitcoindatalabs/ln-client-diversity. All stages truncate their inputs at the data cut-off of 4 October 2026. The underlying gossip and blockchain data are public; our daily gossip snapshots are available from the author on request.

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